Beyond Pass@1: Reliability and Token-Cost Effects of Quantized Reasoning Models under a Pinned Serving Stack

Manish Nandish*
Indian Institute of Technology Patna
PARAM Rudra HPC (NSM / C-DAC)

August 16, 2026

Abstract

Post-training quantization is a common serving default for reasoning language models, but pass@1 on a single seed can hide changes in trace length, termination behavior, and sample-level reliability. We pin one serving stack—vLLM 0.7.0 [10] eager execution on an NVIDIA A100-80GB, with FP8 checkpoints executed as Marlin W8A16 [5] rather than native W8A8—and vary only the weight checkpoint among BF16, FP8, AWQ-4, and GPTQ-4 for DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. The grid has 88 cells and 56,408 completions on MATH-500 [8, 13] (5 seeds), GSM8K [2] (3 seeds), and GPQA-Diamond [16] (3 seeds).

On MATH-500, FP8–BF16 pass@1 differences are +0.40 and +0.28 percentage points (pp). Problem-clustered bootstrap 95% intervals include zero; a ±1 pp TOST equivalence test is *not* passed. Four-bit effects are architecture- and task-dependent: Llama AWQ-4 loses 2.76 pp on MATH-500 (95% CI $[-4.16, -1.44]$, $p < 0.001$) and 1.57 pp on GSM8K; Qwen AWQ-4 loses 5.56 pp on GPQA-Diamond (95% CI $[-9.60, -1.52]$, $p = 0.007$). Majority-vote McNemar tests on maj@5 remain non-significant and do not answer the pass@1 question. Mean MATH-500 length rises 6.3–6.8% for Qwen 4-bit (paired token CIs exclude zero); the increase is concentrated on items where BF16 is correct and the quantized run is not. Identical-word loops are rare but nonzero (25/56,408); none of the rows hit the 32,768 token cap after re-encoding, yet 209 completions sit at $\geq$ 32,500 tokens, and Qwen 4-bit near-cap counts on MATH-500 are about 1.7× BF16. Compact validation JSON does not store extracted answers or `finish_reason`, so we do not report a deployable agreement gate or Guo-style calibration. Cost-of-Pass is a token ranking at a shared 65 tok/s assumption, not measured wall-clock.

Keywords: Reasoning language models, post-training quantization, pinned serving stack, token inflation, selective prediction, Cost-of-Pass.

1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Weight-only post-training quantization (FP8, AWQ-4 [14], GPTQ-4 [4]) is therefore a common serving choice. Isolated pass@1 numbers hide whether compression also changes trace length [12, 15], termination, or the cost of a correct answer [3]. Serving-stack details—engine version, decoding flags, CUDA graphs—can move reasoning scores as much as model changes [9]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment.

We ask four questions under that pin:

*Department of Computer Science & Engineering, Indian Institute of Technology Patna, India. Email: `manishn_iitp@iitp.ac.in`

- **RQ1.** Does quantization change *pass@1* relative to matched BF16, with problem-clustered uncertainty?
- **RQ2.** Do identical-word loops and near-budget terminations occur, and do they differ by format?
- **RQ3.** What can be said about multi-sample reliability without gold labels at serve time?
- **RQ4.** How does output length change a *token-implied* Cost-of-Pass when throughput is held fixed?

Contributions. (1) A released, format-matched, stack-pinned 88-cell grid (56,408 completions) with per-cell JSON. (2) Problem-clustered pass@1 inference showing FP8–BF16 intervals that include zero, a statistically detectable Llama AWQ-4 drop on MATH-500/GSM8K, and a Qwen AWQ-4 drop on GPQA-Diamond—effects that maj@5 McNemar tests miss. (3) Evidence that 4-bit token inflation is real for Qwen on the full MATH-500 grid, is *not* an artifact of a 200-item even-index subset, and is concentrated on format-induced failures rather than uniformly longer correct traces.

2 Related Work

GPTQ [4] and AWQ [14] were originally validated mainly on short-form benchmarks; SmoothQuant [18] addresses activation quantization, which we do not run. Liu et al. [15] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization and report near-lossless W8/W4A16 in many settings, with *no* systematic output lengthening. Lian et al. [12] report the opposite: low-bit PTQ can inflate chain-of-thought length. Li et al. [11] localize 4-bit math failures to early process errors. Alimaskina et al. [1] study loops and delayed commitment under extreme low-bit inference. Our delta is a single pinned vLLM 0.7.0 eager path [10] with five MATH-500 seeds and paired item-level tests, not a new compressor.

Hochlehnert et al. [9] show that seeds, prompts, decoding, and software move reasoning scores; we freeze those factors. Zhong et al. [19] study uncertainty calibration of quantized LLMs using model probabilities—a quantity we do not log. Self-consistency [17] and later unsupervised estimators [20] treat sample agreement as confidence; selective classification [6] supplies the risk–coverage language. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution; we apply it only as a fixed-throughput token proxy.

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM's Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128, `float16`); **GPTQ-4** (group size 128, Marlin). Hugging Face IDs and revisions are listed in the appendix. AWQ-4 checkpoints are third-party community uploads; "Llama AWQ-4 is weak" is a property of those artifacts, not a proof about AWQ as a method.

3.2 Benchmarks and protocol

MATH-500 [8,13] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [2] ($n = 1{,}319$, seeds 42–44) and GPQA-Diamond [16] ($n = 198$, seeds 42–44): breadth. Decoding: zero-shot `<think>` prefix (appendix), $T = 0.6$, top-$p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), max new tokens 32,768. Total: 56,408 completions in 88 cells. MATH and GSM8K are widely used; contamination likely inflates absolute scores but largely cancels in within-model format contrasts.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM `inference.py` path [15] with `gpu_memory_utilization=0.75`. Engine pin: `vLLM==0.7.0`, eager execution. Source files under `configs/models/` are *not* the campaign launcher; they differ in `max_model_len` and KV-cache defaults and should not be read as the effective stack.

3.4 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is a problem-clustered bootstrap of the paired difference $\Delta = \overline{\mathrm{acc}}_{\mathrm{quant}} - \overline{\mathrm{acc}}_{\mathrm{BF16}}$ (resample n items with replacement; retain all seeds per item; $B = 10{,}000$). We report Δ in percentage points, a 95% percentile CI, and a two-sided bootstrap p-value, with Holm–Bonferroni across the six MATH (resp. GSM8K, GPQA) contrasts. A ± 1 pp two one-sided tests (TOST) using the 90% CI is reported as an equivalence check, not as a default claim.

maj@5 McNemar (secondary). A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold. Exact McNemar tests this consensus, not pass@1.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). A row is a *cap hit* if re-encoded completion length is $\geq 32{,}768$. Because decode-then-re-tokenize undershoots the cap (observed max 32,737), we also count *near-cap* rows with $\geq 32{,}500$ tokens. Saved rows do not include vLLM `finish_reason` or output token IDs; phrase-level cycles are invisible to the identical-word detector.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with a clustered bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness.

Token-implied Cost-of-Pass. Following Erol et al. [3], $C_{\mathrm{pass}} = (\text{cost per query})/\text{pass@1}$ with

$$\text{Cost per query} = (\bar{T}/\tau) \times (R/3600), \tag{1}$$

$\tau = 65$ tok/s (unmeasured) and $R = \$1.50$ per A100-hour. Because τ and R are shared, ranking by C_{pass} is ranking by $\bar{T}/\text{pass@1}$. Measured 4-bit kernel speedups can reverse it.

What we do not report as operational. Compact JSON stores `extractive_match` but not extracted answer strings. A deployable self-consistency gate needs modal-answer agreement [6, 17] *without* gold. Gold-hit ECE (confidence $= c_i/5$, label $= \mathbb{I}(c_i \geq 3)$) is a deterministic function of the histogram of c_i and is omitted from the result tables.

4 Results

4.1 MATH-500 pass@1

Table 1 and Figure 4 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. Llama AWQ-4 is the weak MATH cell (−2.76 pp).

Table 1: MATH-500 headline grid ($n = 500$, seeds 42–46). Clustered 95% CIs resample problems. Loops: identical-word run ≥ 20. Near-cap: completion tokens $\geq$ 32,500. Exact cap hits after re-encoding: 0 in all cells.

Cell	**42**	**43**	**44**	**45**	**46**	**Mean ± Std**	**Clust. 95%**	**Tok.**	**L**	**NC**
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	2	21

Table 2: Primary: problem-clustered bootstrap of MATH-500 pass@1 versus BF16 ($B =$ 10,000). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts. TOST ± 1 pp uses the 90% CI.

Contrast	Δ **(pp)**	**95% CI**	p	**Holm**	**McNemar** p
Llama AWQ-4	-2.76	$[-4.16, -1.44]$	< 0.001	sig.	0.296
Qwen AWQ-4	-0.88	$[-1.88, +0.08]$	0.069	n.s.	1.000
Qwen GPTQ-4	-0.52	$[-1.40, +0.36]$	0.245	n.s.	0.754
Qwen FP8	$+0.40$	$[-0.48, +1.28]$	0.383	n.s.	0.581
Llama GPTQ-4	-0.32	$[-1.56, +0.96]$	0.619	n.s.	1.000
Llama FP8	$+0.28$	$[-0.92, +1.48]$	0.647	n.s.	1.000

Table 2 is the test that matches RQ1. Llama AWQ-4's clustered interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). Qwen 4-bit MATH drops are negative but not Holm-significant. No MATH contrast, including FP8, has its 90% CI inside $[-1, +1]$ pp, so we do *not* claim statistical equivalence at a 1 pp serving margin. Secondary maj@5 McNemar p-values are all > 0.29: majority voting absorbs the per-seed flips that pass@1 still sees.

4.2 GSM8K and GPQA-Diamond

Table 5: FP8 stays within about one point of BF16 on all three tasks. The largest accuracy effect in the study is Qwen AWQ-4 on GPQA-Diamond (-5.56 pp, 95% CI $[-9.60, -1.52]$, $p = 0.007$, Holm-significant among six GPQA contrasts; 3 seeds, $n = 198$). Llama AWQ-4 also drops on GSM8K (-1.57 pp, 95% CI $[-2.55, -0.58]$, $p = 0.002$). Other GSM8K Qwen contrasts pass TOST at ± 1 pp. GPQA intervals are wide; we treat non-AWQ GPQA contrasts as exploratory.

4.3 Loops and near-cap termination

Across 88 cells the validator records **25** repetition-flagged rows (0.044% of 56,408) and **0** exact cap hits. Earlier drafts reported 0/0 because analysis scripts read `truncation_count` / `repetition_flag_count` while the validator writes `token_limit_hits` / `repetition_rows`. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ). Three runs exceed 10,000 identical consecutive words. Near-cap rows ($\geq$ 32,500 tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16's 14, and almost all near-cap rows are unboxed. Without `finish_reason` these are inferred length terminations, not proven `length` stops.

Table 3: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ: mean paired per-item token difference with clustered 95% CI. "Both OK" / "BF16 only": mean Δ on (item, seed) pairs with that correctness pattern.

Contrast	**RoM**	**Mean Δ [95% CI]**	**Both OK**	**BF16 only**
Qwen FP8	−0.1%	−4 [−142, +134]	+51	+5444
Qwen AWQ-4	+6.3%	+254 [+116, +398]	+166	+6915
Qwen GPTQ-4	+6.9%	+276 [+121, +431]	+136	+6892
Llama FP8	−2.3%	−106 [−284, +65]	+38	+2629
Llama AWQ-4	+1.7%	+80 [−109, +268]	+91	+3090
Llama GPTQ-4	+3.9%	+184 [+12, +359]	+66	+5915

Table 4: Token-implied Cost-of-Pass on MATH-500 at $1.50/A100-hour and a shared 65 tok/s. Not measured wall-clock. Ranking equals ranking by mean tokens / pass@1.

Cell	**Mean tok.**	**Cost / Q**	C_{pass}
Qwen BF16	4,011.4	$0.02571	$0.02736
Qwen FP8	4,007.7	$0.02569	$0.02721
Qwen AWQ-4	4,265.3	$0.02734	$0.02936
Qwen GPTQ-4	4,287.2	$0.02748	$0.02940
Llama BF16	4,656.6	$0.02985	$0.03345
Llama FP8	4,550.8	$0.02917	$0.03259
Llama AWQ-4	4,736.5	$0.03036	$0.03511
Llama GPTQ-4	4,840.5	$0.03103	$0.03490

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 3, Figure 2), Qwen AWQ-4/GPTQ-4 generate +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A previous 200-item "stratified mixed-correctness" analysis used even indices among the first 400 problems, seed 42 only, and the *mean of per-item ratios*. That estimator is not the full-grid ratio of means: on the same 200 items it reports Qwen FP8 +11.4% while the ratio of means is −3.4%. We discard that subset interpretation.

Figure 3 and Table 3 split paired token deltas by correctness. When both formats are correct, 4-bit adds on the order of +90 to +170 tokens. When BF16 is correct and the quantized sample is wrong, the quantized trace is thousands of tokens longer. Token inflation is therefore largely a *failure-associated* lengthening, which is a more useful reconciliation of Liu et al. [15] (little mean lengthening) and Lian et al. [12] (hidden token cost) than a 200-item even-index slice.

Under the shared-τ model (Table 4, Figure 1), FP8 has the lowest token-implied C_{pass} by $0.00015 on Qwen. That margin is smaller than seed noise in length. We do not call it Pareto-optimal.

4.5 Difficulty labels

Table 6: Level 1 is easy for every format in this sample. Llama AWQ-4 is weakest on Levels 2–4. MATH level labels are not a monotone difficulty axis here (Level 2 Qwen BF16 89.11% vs Level 5 92.69%).

Table 5: Cross-benchmark pass@1 (mean ± seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

Cell	**MATH-500**	**GSM8K**	**GPQA-D**	**L / NC**
Qwen BF16	94.00 ± 0.55	91.26 ± 0.29	50.34 ± 2.96	4 / 28
Qwen FP8	94.40 ± 1.05	91.33 ± 0.16	49.49 ± 1.52	0 / 24
Qwen AWQ-4	93.12 ± 0.67	91.05 ± 1.14	44.78 ± 3.04	5 / 32
Qwen GPTQ-4	93.48 ± 0.77	91.13 ± 0.27	47.98 ± 1.75	3 / 40
Llama BF16	89.24 ± 0.74	88.68 ± 0.46	46.13 ± 1.91	3 / 21
Llama FP8	89.52 ± 1.01	88.80 ± 0.62	47.81 ± 0.29	3 / 12
Llama AWQ-4	86.48 ± 1.96	87.11 ± 0.23	46.97 ± 2.02	4 / 21
Llama GPTQ-4	88.92 ± 1.55	88.96 ± 0.70	44.95 ± 4.32	3 / 31

Table 6: MATH-500 difficulty labels (5-seed pass@1 mean ± std). Labels are not monotone: Level 2 is harder than Level 5 for both BF16 backbones. Level-1 $n = 43$; treat cell-wise gaps as descriptive.

Level	**Qwen BF16**	**Qwen FP8**	**Qwen AWQ-4**	**Qwen GPTQ-4**
1 ($n = 43$)	98.60 ± 1.27	98.60 ± 2.08	98.14 ± 1.95	96.28 ± 2.08
2 ($n = 90$)	89.11 ± 0.93	90.44 ± 4.20	90.67 ± 1.69	88.67 ± 2.14
3 ($n = 105$)	95.81 ± 1.09	95.24 ± 1.51	94.10 ± 0.80	94.29 ± 1.90
4 ($n = 128$)	95.78 ± 0.89	96.41 ± 1.42	94.53 ± 1.24	96.56 ± 1.18
5 ($n = 134$)	92.69 ± 1.11	93.13 ± 1.23	91.04 ± 1.67	92.24 ± 1.95
Level	**Llama BF16**	**Llama FP8**	**Llama AWQ-4**	**Llama GPTQ-4**
1 ($n = 43$)	93.95 ± 3.12	94.88 ± 3.03	95.35 ± 1.64	96.28 ± 2.65
2 ($n = 90$)	84.67 ± 2.14	85.11 ± 2.02	81.11 ± 3.04	83.33 ± 3.42
3 ($n = 105$)	89.90 ± 2.48	90.10 ± 1.73	85.71 ± 1.17	89.71 ± 1.83
4 ($n = 128$)	91.56 ± 1.69	91.09 ± 0.43	87.03 ± 2.25	89.84 ± 1.75
5 ($n = 134$)	88.06 ± 1.75	88.81 ± 2.84	87.31 ± 3.46	88.81 ± 1.18

4.6 Multi-sample reliability, without a safety gate

Of 500 MATH-500 items, Qwen FP8 has 444 items with all five seeds gold-correct and 8 with all five gold-wrong. Those counts are properties of gold match, not of whether the five extracted strings agree. Compact JSON has no answer strings, so we cannot recompute a modal-agreement risk–coverage curve. We therefore do not report $k/5$ "agreement" operating points and do not call sample-consistency an operational safety gate. Recovering answers from the original HPC JSONLs would make that analysis deployable.

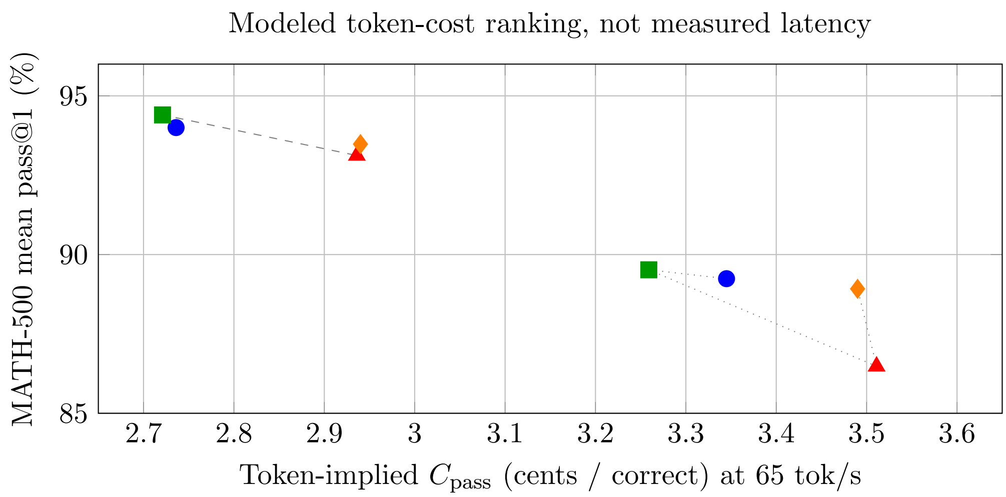

Figure 1: Token-implied C_{pass} versus pass@1 (cents). Solid/dashed lines are Qwen/Llama. Throughput is assumed equal, so the plot is a re-parameterization of Table 1.

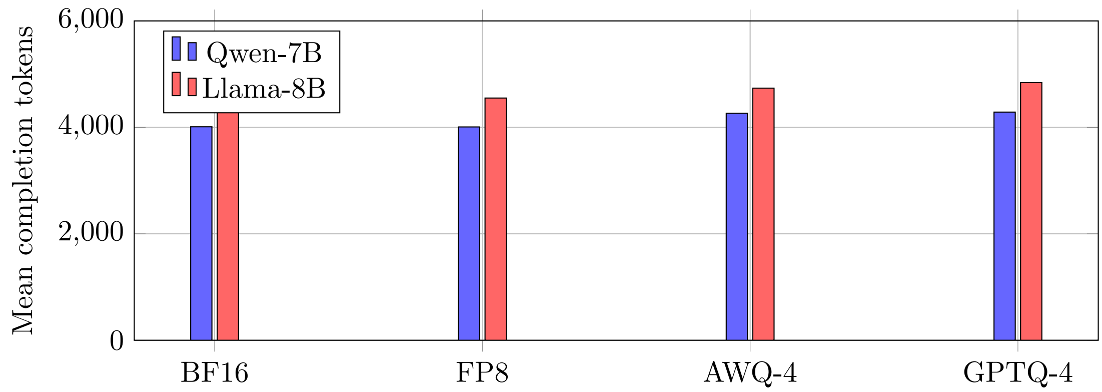

Figure 2: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand).

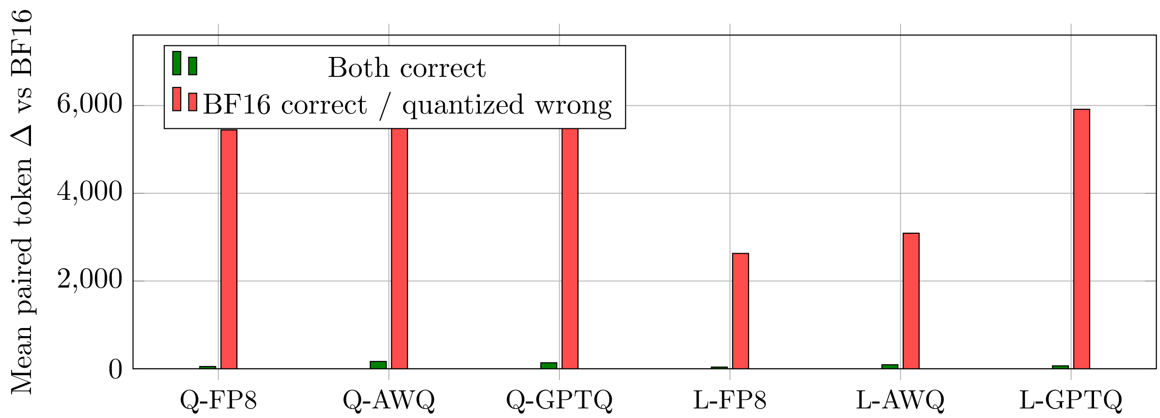

Figure 3: Paired token deltas by correctness. Extra length is concentrated on format-induced failures, not on uniformly longer correct traces.

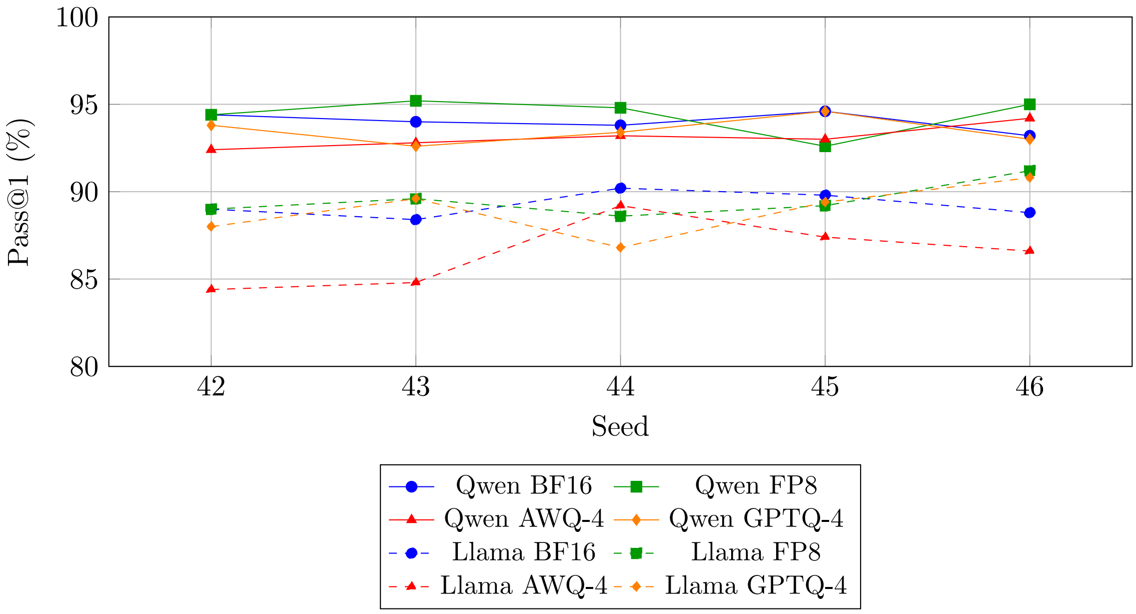

Figure 4: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. Llama AWQ-4 is the unstable 4-bit cell.

5 Discussion

Under this pinned A100 W8A16 stack, FP8-checkpoint serving tracks BF16 pass@1 closely on the evaluated distillations, but we do not establish ± 1 pp equivalence on MATH-500. Four-bit behavior is not a single story: Qwen keeps MATH-500 and GSM8K near BF16 while losing 5.6 pp on GPQA-Diamond under AWQ-4; Llama AWQ-4 is the MATH/GSM8K weak cell. Mean 4-bit lengthening on MATH-500 is modest and, when present, lives mostly on traces that also lose the answer. That is a sharper claim than "4-bit inflates reasoning" and a more honest one than "nothing happens."

We previously wrote that pinning vLLM 0.7.0 "removed" truncations and loops seen on vLLM 0.8.5. Those earlier runs also differed in harness, decoding flags (including a bug that dropped `repetition_penalty`), and prompt profile. The final grid still contains 25 loops. We therefore do not attribute pathology rates causally to the engine version. A clean stack $\times$ format factorial remains future work, not a result in this paper.

FP8 is a reasonable *accuracy* default on this A100 W8A16 path for these two distillations. It is not a measured speed default.

6 Limitations

1. **A100 W8A16 fallback.** FP8 numbers do not transfer to Hopper native W8A8.

2. **Unmeasured serving cost.** C_{pass} assumes equal 65 tok/s. Weight-only 4-bit kernels typically raise decode throughput; the ranking can flip. Peak VRAM, energy, and GPU-hours are not logged per cell. At 32k traces the KV cache can dominate weight memory; weight-only PTQ does not shrink that term.

3. **Pathology metrology.** The loop detector is identical-word runs ≥ 20. Phrase-level cycles and `finish_reason=length` are unobserved. Near-cap is a proxy.

4. **No extracted answers in the public JSON.** Modal-agreement selective prediction and non-circular ECE cannot be recomputed from the release. Gold-hit gates are not deployable.

5. **Checkpoint provenance.** AWQ-4 is a community quantization of unknown calibration set. GPTQ-4/FP8 are RedHatAI artifacts (appendix).

6. **Two 7–8B distillations**, English math/science, one engine, one GPU generation, one prompt. GSM8K/GPQA use three seeds.

7. **Source configs vs. launcher.** Files in `configs/models/` are not the QRM `inference.py` campaign. Effective KV-cache dtype is not stored per cell.

7 Conclusion

On a pinned vLLM 0.7.0 eager stack, FP8-checkpoint (W8A16) pass@1 stays close to BF16 for two R1 distillations, while 4-bit effects are model- and task-dependent and are visible to clustered pass@1 tests that maj@5 McNemar misses. Loops are rare but nonzero; near-cap terminations are more common for Qwen 4-bit on MATH-500. Mean token inflation, where it exists, is concentrated on failures. Token-implied Cost-of-Pass at a shared throughput is a length ranking, not a measured frontier. The contribution is a stack-pinned measurement of those effects, not a new compressor and not a serving-stack shift.

Artifacts

Code, patches, validation JSON, and analysis reports: `https://github.com/Manish06N/reasoning-compress` Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Corrected tables: `results/reports/revision_reanalysis_report.json`. JSON rows contain `extractive_match`, `completion_tokens`, `repetition_flag`; they do *not* contain traces, token IDs, extracted answers, or `finish_reason`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

Acknowledgment

PARAM Rudra HPC at IIT Patna (National Supercomputing Mission, C-DAC) provided compute.

A Prompt

MATH-500 and the GSM8K/GPQA QRM templates use:

```
{question}

Please reason step by step, and put your final
answer within \boxed{}.
```

with a <think> prefix on the official QRM path.

B Checkpoints

Cell	Hugging Face ID	Revision
Qwen BF16	`deepseek-ai/DeepSeek-R1-Distill-Qwen-7B`	`916b56a4...`
Qwen FP8	`RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic`	`ceb2fcd1...`
Qwen AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ`	`1d082b3d...`
Qwen GPTQ-4	`RedHatAI/...-quantized.w4a16`	`9559a91e...`
Llama BF16	`deepseek-ai/DeepSeek-R1-Distill-Llama-8B`	`6a6f4aa4...`
Llama FP8	`RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic`	`5d548d91...`
Llama AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ`	`57360e84...`
Llama GPTQ-4	`RedHatAI/...-quantized.w4a16`	`faa06ed0...`

C Reanalysis

Pathology, clustered bootstrap, and token strata are produced by `scripts/analysis/revision_reanalysis.p` from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10{,}000$, seed 0.

References

[1] Ekaterina Alimaskina, Darya Rudas, Denis Shveykin, Gleb Molodtsov, Pavel Vasiliev, and Aleksandr Beznosikov. Extreme low-bit inference in reasoning models: Failure modes and targeted recovery. *arXiv preprint arXiv:2606.02011*, 2026.

[2] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[3] Mehmet Hamza Erol, Batu El, Mirac Suzgun, Mert Yüksekgönül, and James Zou. Cost-of-pass: An economic framework for evaluating language models. *arXiv preprint arXiv:2504.13359*, 2025.

[4] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.

[5] Elias Frantar, Roberto L. Castro, Jiale Chen, Torsten Hoefler, and Dan Alistarh. MARLIN: Mixed-precision auto-regressive parallel inference on large language models. *arXiv preprint arXiv:2408.11743*, 2024.

[6] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

[7] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

[8] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Neural Information Processing Systems (NeurIPS)*, 2021.

[9] Andreas Hochlehnert, Hardik Bhatnagar, Vishaal Udandarao, Samuel Albanie, Ameya Prabhu, and Matthias Bethge. A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility. *arXiv preprint arXiv:2504.07086*, 2025.

[10] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

[11] Zhen Li, Yupeng Su, Songmiao Wang, Runming Yang, Congkai Xie, Aofan Liu, Ming Li, Jiannong Cao, Yuan Xie, Ngai Wong, and Hongxia Yang. Quantization meets reasoning: Exploring and mitigating degradation of low-bit LLMs in mathematical reasoning. *arXiv preprint arXiv:2505.11574*, 2025.

[12] Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. Quantization inflates reasoning: Token inflation as a hidden cost of low-bit reasoning models. *arXiv preprint arXiv:2606.25519*, 2026.

[13] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

[14] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of the Conference on Machine Learning and Systems (MLSys)*, 2024.

[15] Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. Quantization hurts reasoning? an empirical study on quantized reasoning models. *arXiv preprint arXiv:2504.04823*, 2025.

[16] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*, 2023.

[17] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[18] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning (ICML)*, pages 38087–38099, 2023.

[19] Mingyu Zhong, Guanchu Wang, Yu-Neng Chuang, and Na Zou. Quantized can still be calibrated: A unified framework to calibration in quantized large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 30503–30517, 2025.

[20] Thomas Zollo, Jimmy Wang, and Richard Zemel. Unsupervised confidence calibration for reasoning LLMs from a single generation. *arXiv preprint arXiv:2604.19444*, 2026.